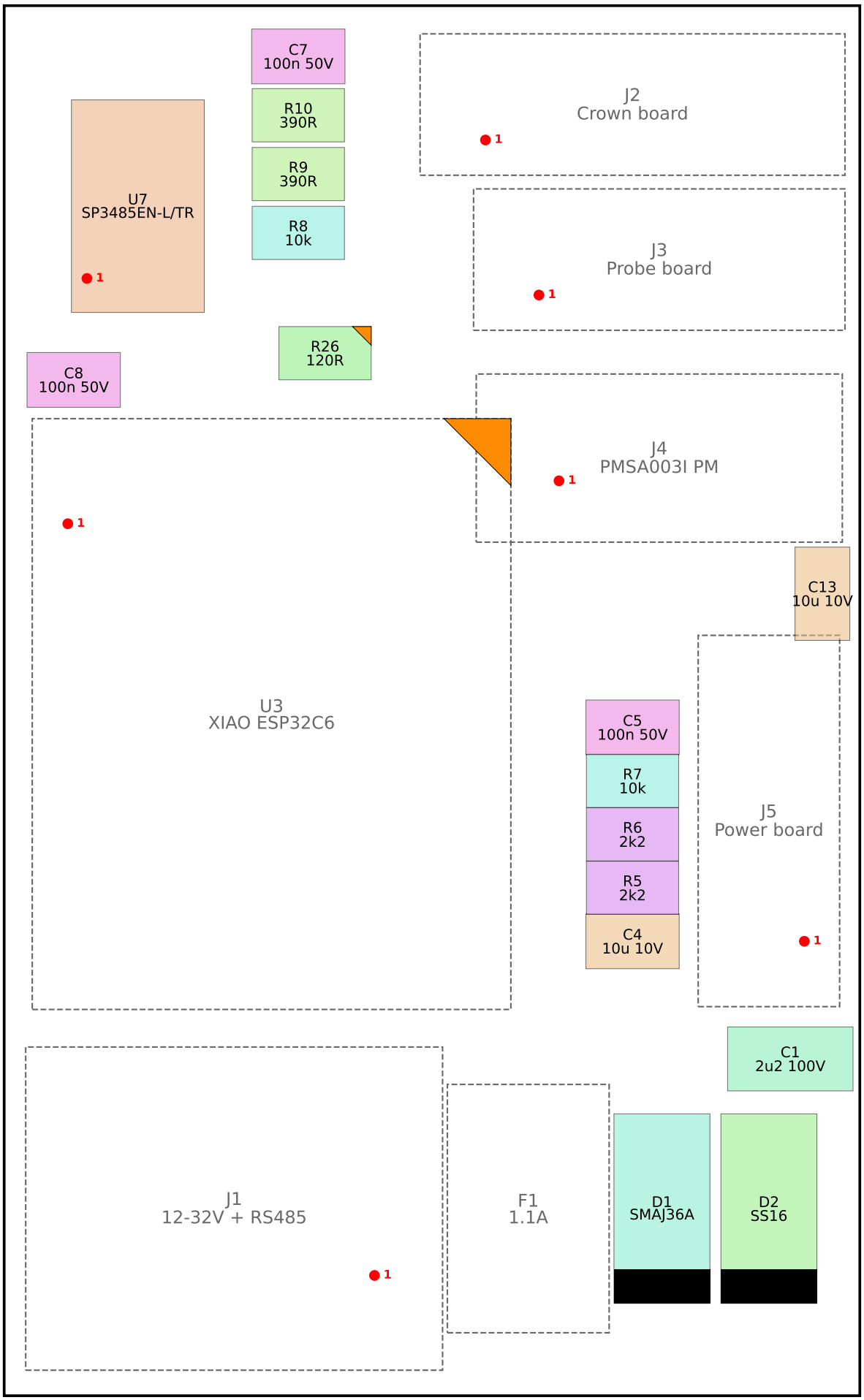
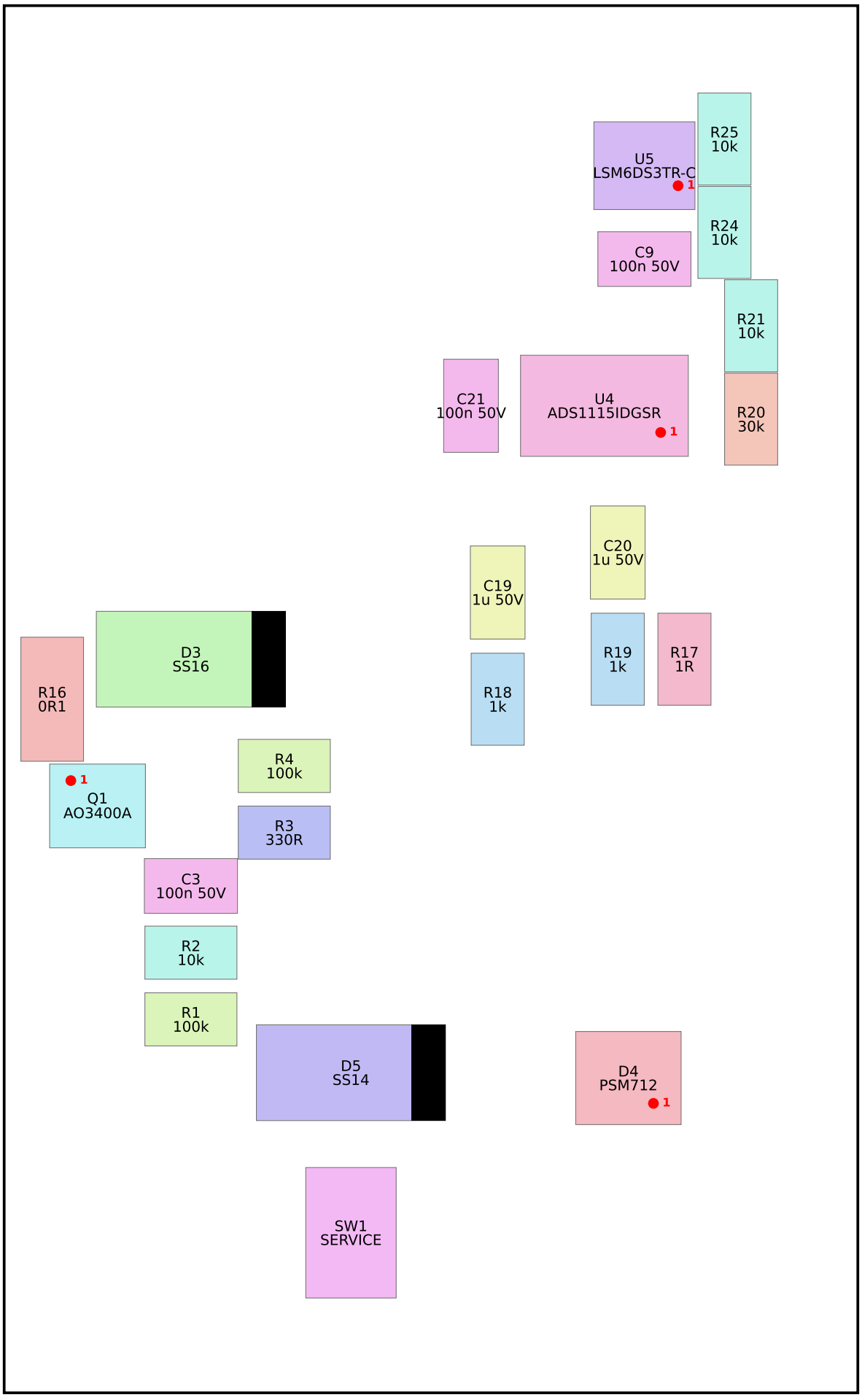
main - TOP

red dot 1 = pin 1 black bar = cathode red + = cap + orange corner = hand-fitted dashed = fit last (key: last page)



main - BOTTOM (seen from below, mirrored)

red dot 1 = pin 1 black bar = cathode red + = cap + orange corner = hand-fitted dashed = fit last (key: last page)



main - values

- 0R1 x1 bottom: R16
- 1.1A x1 top: F1
- 100k x2 bottom: R1, R4
- 100n 50V x3 top: C5, C7, C8
- 100n 50V x3 bottom: C3, C9, C21
- 10k x2 top: R7, R8
- 10k x4 bottom: R2, R21, R24, R25
- 10u 10V x2 top: C4, C13
- 12-32V + RS485 x1 top: J1
- 120R x1 top: R26
- 1R x1 bottom: R17
- 1k x2 bottom: R18, R19
- 1u 50V x2 bottom: C19, C20
- 2k2 x2 top: R5, R6
- 2u2 100V x1 top: C1
- 30k x1 bottom: R20
- 330R x1 bottom: R3
- 390R x2 top: R9, R10
- ADS1115IDGSR x1 bottom: U4
- AO3400A x1 bottom: Q1
- Crown board x1 top: J2
- LSM6DS3TR-C x1 bottom: U5
- PMSA003I PM x1 top: J4
- PSM712 x1 bottom: D4
- Power board x1 top: J5
- Probe board x1 top: J3
- SERVICE x1 bottom: SW1
- SMAJ36A x1 top: D1
- SP3485EN-L/TR x1 top: U7
- SS14 x1 bottom: D5
- SS16 x1 top: D2
- SS16 x1 bottom: D3
- XIAO ESP32C6 x1 top: U3

main - key

Marks on the board pages

red dot + 1 : pin 1 of ICs, transistors, LEDs, optocouplers, connectors and modules

black bar : diode cathode, matching the band printed on the diode body

red + : + pad of electrolytic / tantalum capacitors

no mark : resistors, ceramic capacitors, inductors, fuses, crystals, pin strips, bidirectional TVS

orange corner: part fitted by hand (not placed by JLCPCB); dashed outline: large part, fit it last

the BOTTOM page is mirrored: it shows the board as you see it from below

Finding pin 1 on the real part

SOT-23 (3 legs): the single leg on its own side is pin 3; pin 1 is left of it on the two-leg side

SOT-23-5 / SC-70 / SOIC / VSSOP / QFN / LGA: a dot, dimple or bevelled edge marks pin 1

Addressable LEDs: a marked (cut / green) corner - check the datasheet, it is easy to misread

Diodes: the band is the cathode (K)

Electrolytic capacitors: the stripe marks the - side (the other pad is +)

Tools

0805 / SOT-23: a 3-5x magnifier or a 10x loupe

QFN / LGA / DFN / SC-70: a USB microscope (check the pads under the part after reflow)